



ASSESSMENT GUIDE

COMM5000

Data Literacy

Case Study Project
Milestone 1 Information

Term 3, 2026

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Assessment Summary

Assessment Task	Weighting	Due Date*	Course Learning Outcomes
Milestone 1: Case Study Preliminary Insight Development	20%	Week 4	1, 2
Milestone 1: Peer assessment	5%	Week 6	
Milestone 1: Evaluation of Peer assessment		Week 7	
Milestone 2: Case Study project proposal: hypothesis tests	25%	Week 7	1, 2, 3, 4
<i>Smarthinking/Studiosity/Feedback Hub feedback (formative task)</i>	0%	Week 9	
Case Study business report	50%	Week 10	2, 3, 4, 5
<i>* Due dates are set at Australian Eastern Standard/Daylight Time (AEST/AEDT). If you are located in a different time-zone, you can use the time and date converter.</i>			

Assessment Administrative Details

Turnitin

Turnitin is an originality checking and plagiarism prevention tool that enables checking of submitted written work for improper citation or misappropriated content. Each Turnitin assignment is checked against other students' work, the Internet and key resources selected by your Course Coordinator.

If you are instructed to submit your assessment via Turnitin, you will find the link to the Turnitin submission in your Moodle course site. You can submit your assessment well before the deadline and use the Similarity Report to improve your academic writing skills before submitting your final version.

You can find out more information on the [Turnitin information site for students](#).

Late Submissions

The parameters for late submissions are outlined in the [UNSW Assessment Implementation Procedure](#). For COMM5000, if you submit your assessments after the due date, you will incur penalties for late submission unless you have Special Consideration (see below). Late submission is 5% per day (including weekends), calculated from the marks allocated to that assessment (not your grade). Assessments will not be accepted more than 5 days late.

Extensions

You are expected to manage your time to meet assessment due dates. If you do require an extension to your assessment, please make a request as early as possible before the due date via the special consideration portal on [myUNSW](#) (My Student profile > Special Consideration). You can find more information on Special Consideration and the application process below. Lecturers and tutors do not have the ability to grant extensions.

Smarthinking/Studiosity/Feedback Hub

The [Feedback Hub at UNSW](#) is a valuable resource for students looking to improve their writing skills. It offers personalized feedback on drafts of essays, reports, and literature reviews, focusing on structure, grammar, referencing, and language choice. The service is free for all UNSW students and is available 24/7, ensuring help is always accessible. Students can expect comprehensive feedback within 24 hours, making it a quick and efficient way to enhance their writing. For more details, visit the Feedback Hub website.

Please keep the following in mind when submitting your drafts to Studiosity:

- Make sure you click on the Studiosity link in the Moodle course (circled below in red) and NOT on the side bar (highlighted in yellow)
- If you receive an error message indicating that Studiosity is unable to connect, please use a different browser from the one that you are using.
- If you have any technical issues, please contact Studiosity directly on support@studiosity.com or for a faster response, visit their website: <https://www.studiosity.com/> and use the chat box (yellow icon on the bottom right) to report your issue.

Special Consideration

Special consideration is the process for assessing the impact of short-term events beyond your control (exceptional circumstances), on your performance in a specific assessment task.

What are circumstances beyond my control?

These are exceptional circumstances or situations that may:

- Prevent you from completing a course requirement,
- Keep you from attending an assessment,
- Stop you from submitting an assessment,
- Significantly affect your assessment performance.

Available here is a [list of circumstances](#) that may be beyond your control. This is only a list of examples, and your exact circumstances may not be listed.

You can find more detail and the application form on the [Special Consideration](#) site, or in the UNSW Special Consideration Application and Assessment Information for Students.

Case Study Information

Business context: India's used car market has grown rapidly in recent years and now sells more vehicles each year than the new car market. As a result, online marketplaces and dealer groups are investing in new technologies to enhance both vehicle pricing and sales. Accurate price appraisal plays a vital role in these processes and currently relies heavily on manual valuations by dealers and inspection experts.

Case/Scenario: You consult for an online used car marketplace operating across a number of major Indian cities, and you help this company to predict or estimate the market price of a used car at the point of appraisal. Knowing the expected price will allow the marketplace to be better positioned to make instant cash offers, flag under- and over-priced listings, and predict inventory turnover and yearly sales. It will also support the appraisal teams by potentially improving the quality and speed of their decisions. Furthermore, similar techniques can help in target marketing by modelling buyer preferences in niche market segments. In order to predict used car prices you will use a dataset consisting of 1,005,000 used car listings covering several brands across a number of Indian cities, and the statistical methods covered in this course.

Project objectives

1. Looking at the provided dataset, is there any relationship (positive or negative) that can be used between the price of a used car and any of the variables in the dataset? Does brand segment (luxury vs non-luxury) have an impact on our predictions of used car prices? If so, how can these relationships be used to predict used car prices, using the methods used in this class? For this project, the luxury segment is defined as the brands Audi, BMW and Mercedes; all other brands form the non-luxury segment. The data provided is everything we know about these cars, and no external data sources are to be used. Moreover, no knowledge about the automotive industry is assumed or required: this project should be seen as a business / statistical exercise. Unlike the datasets used in the seminars, this dataset contains realistic imperfections: inconsistent category labels, 'Unknown' entries, missing values, duplicated records, and some implausible extreme values. Part of your task is to detect these issues, decide how to deal with them, and clearly document and justify your choices. Please provide both quantitative and qualitative analysis supporting any findings.
2. In addition, based on your analysis towards the Objective (1) identify weaknesses and limitations of the chosen approach, and propose, at least in broad terms, a better approach. This proposed approach could include additional data to be included, or a methodology that is able to better deal with data limitations. Please also provide any supporting analysis for these additional considerations.

COMM5000 Context

This is a business question that is based on real data, although simulated for assessment purposes. The role you are to play is one of a consultant contracted by an online used car marketplace to assist with the analysis of data using the COMM5000 data analysis tools, which include descriptive and inferential statistics.

The work will be scaffolded into two milestones M1 (25%) & M2 (25%) and a final project report (50%). Every milestone will require you to use what you have learned to address specific aspects of the data. Generally, M1 consists of an exploratory data analysis, whereas M2 is concerned with identifying hypotheses and formulating key inferential questions. In the final project report, all the insights gathered from M1 and M2 are used to model the data to answer the project questions.

M1 is a peer-reviewed assessment, which means that your assessment will be assessed by some random peer students. More details on this process below.

Schedule of engagement for the entire course

Upon request and as part of additional support for assessments one of the course teaching team members might hold consultation sessions throughout the term. It is very important that you attend these sessions where we

will hold live synchronous sessions to provide more detailed information about the case study. During these sessions, you are free to ask questions and discuss any aspects of the project.

Milestone 1: Preliminary Insight Development

Report details



Week 4, Friday, at 5PM



25% (20% +5%)



Report: This is individual work. Reports will be checked for plagiarism.



1000 words (not including tables, graphs, and references)



Via Moodle course site

Description of assessment task

This first milestone aims to give you a better understanding of the datasets, variables, and questions in this Case Study. This exploratory data analysis seeks to get the necessary insights so that a development plan can be formulated to address the following key points of the case study project:

- 1) Data analysis: an in-depth description of the variables included in the dataset and the relationship between used car prices and vehicle age (Registration_Age), as well as between used car prices and usage (Kms_Driven).
- 2) Effect of Brand Segment (luxury vs non-luxury) on estimated used car prices.

You must submit a written development plan summarising the finding from the data explorations, describing any patterns from comparing summary statistics of the variable of interest, and providing a plan on how you may address the key questions (1) and (2). The report should be concise and well written.

Please note that you are not required to fully answer (1) and (2) in this milestone. Instead, you are required to develop insights and understand the problem, as well as the datasets for your final project.

As a style guide, you may include some or all tables/graphs as an appendix and refer to them as appropriate in your report. You should only include graphics and tables to support your analysis, conclusions, and findings. While preparing your paper, you will encounter numerous tables and graphs, which are irrelevant to the analysis. So be very selective and make good use of the page limit!

Approach to the assessment task

In week 1, we learnt how to represent the data using graphical tools, as well as numerical summaries. All these tools are meant to give us an idea of what the data are 'trying to tell us'. Can we make sense of the large numbers of observations and tell a simple story or pick up a trend? This is what you will do in this milestone: understand the data and what we are trying to find out from the data.

(A) Expected Tasks

- (i) Download the data. This assessment requires the download of the Excel file provided on your course Moodle page (file name: "COMM5000-Used_Car_Price_Prediction.xlsm"). The dataset contains 1,005,000 used car listings. A description of every variable is provided on the first worksheet of the Excel file.
- (ii) Data preparation. Unlike the datasets used in the seminars, these data are NOT fully clean. The dataset contains

inconsistent category labels (for example, ' BMW ' with extra spaces, or 'PETROL', 'petrol' and 'hybrid' as variants of existing fuel types), 'Unknown' entries, missing values in some numeric variables, a small number of duplicated records, and some implausible extreme values (for example, cars with more than 1,000,000 kilometres driven). You must identify these issues, explain any data manipulations you perform, and provide a rationale for them. Hint: Excel functions such as TRIM and LOWER, and tools such as PivotTables, filters and conditional formatting, are useful for auditing categorical variables.

- (iii) **Variables of interest.** Consider (1) and (2) above and focus only on the variables included in the provided dataset.

(B) The expected outcomes

The written work must provide a brief description of the Case Study problem and a clear plan of how the dataset provided will address the key questions raised in the project description. You will have the opportunity to adjust, revise and review this plan as we progress throughout the term. M1 analysis is based on COMM5000 content covered in weeks 1, 2 and 3.

- (i) Numerical summaries of the key variables of interest: present descriptive statistics of the variables in the data. You may represent these results in the form of tables.
- (ii) These numerical summaries must be presented for 1) the entire sample, 2) only for luxury cars (Audi, BMW, Mercedes) and 3) only for non-luxury cars (all other brands). Be careful: the Brand labels must be cleaned before the two segments can be constructed correctly.
- (iii) For example, for each variable:

	Mean	Mode	Median	SD	Min	Max
Variable name 1						
...						

- (ii) Graphical representations of some variables if you deem it important for to capture a trend or some interesting patterns in the data.
- (iii) Analysis of the relationship between used car prices and vehicle age (Registration_Age), and between used car prices and kilometres driven (Kms_Driven). Use scatter plots and describe your findings. Practical note: with 1,005,000 observations, plotting every point is neither feasible nor informative in Excel. You may build your scatter plots on a random sample of observations (e.g., 2,000–5,000 rows) or plot group means (e.g., average price by vehicle age). Explain and justify the approach you take.
- (iv) What conclusions can you make from the inspection of these data summaries in the form of tables and graphs? For example, is there a pattern that you can identify? What does the difference between the mean and the median of Car_Price tell you about the shape of the price distribution, and about which summary statistic you should report to the client?
- (v) Your analysis should inform your development plan to address points (1) and (2) in Milestone 2 and in the final report. This plan may be revised later during your work on Milestone 2.

Structure of the report

* **Introduction** You should briefly introduce the topic and summarise the purpose and importance of this project for the client. Then outline how this preliminary insight development plan will be structured. It is important to provide some background information on this topic. You can find relevant information from the dataset publisher's page [Stable Space, 2026]: <https://www.kaggle.com/datasets/sharmajicoder/used-car-price-prediction-dataset> , or from publicly available reports on the Indian used car market.

* **Data Summaries and Descriptive Statistics:** Provide the necessary analysis to explore the variables. Describe the trends and stories that emerge from the data summaries. Are any patterns emerging from the graphs or tables you have constructed so far? Note: now that you have completed the first stage of data summaries, you have some basic insight into the dataset. You can use this information to develop some plans of action to address points (1) and (2) in Milestone 2 and in the final report.

* **Conclusion:** The conclusion should summarise the findings of your investigation and any concluding comments. It should also provide your plan for the next step of the analysis.

* **References:** Every piece of external documents you use needs to be properly referenced. Please include page and link so that your lecturer and tutor can efficiently check your work.

* It is suggested that you limit your report to a maximum of 8 pages including tables, graphs, and references.

Schedule of engagement for M1

Below you can find a summary of the deadlines related to M1. M1 requires not only the submission of the report (15%), but also a high-quality, constructive contribution to the assessment of the other students' works (5%). This peer assessment involves being randomly assigned to critically review a number of other students' submissions. You cannot

choose which students to assess. You will 1) mark these submissions using a marking rubric to facilitate your job but also 2) leave constructive feedback. It is important that you leave high-quality, constructive feedback. Your feedback will be assessed, and you will also have the opportunity to evaluate the feedback you have received.

1. **Week 4**, Friday, deadline for submission of M1
2. **Week 6**, deadline for marking the submissions you were assigned to
3. **Week 7**, deadline for evaluating the feedback you have received

Submission instructions

- Via Moodle course site.

Supporting resources and links

- **Dataset files:** The Excel dataset file is available on Moodle. You only need to analyse the data that is included in this file.
- **Weekly seminar:** The seminar coordinator will cover relevant project aspects using Excel during the SEM session.
- **Background information:** See the details included in the provided dataset. This dataset was retrieved at the provided location in June 2026, and was removed from the original repository at some point after this date.

Assessment AI Mode

Simple Editing Assistance

In completing this assessment, you are permitted to use standard editing and referencing functions in the software you use to complete your assessment. These functions are described below. You must not use any functions that generate or paraphrase passages of text or other media, whether based on your own work or not.

If your Convenor has concerns that your submission contains passages of AI-generated text or media, you may be asked to account for your work. If you are unable to satisfactorily demonstrate your understanding of your submission you may be referred to UNSW Conduct & Integrity Office for investigation for academic misconduct and possible penalties.

For more information on Generative AI and permitted use please see [here](#).

Milestone 1 Marking Rubrics

Criteria	Ratings					Pts
Business context and articulation of problem(s) (10%) Task(i) (See Milestone 1 information)	High Distinction (85-100%) The problem and business context are clearly and correctly identified and defined, providing significant original, innovative, advanced insight into the problem.	Distinction (75-84%) The problem and business context are clearly and considerably identified and defined, providing some original, innovative, advanced insight into the problem.	Credit (65-74%) The problem and business context are mostly identified and defined, providing some original, basic insight into the problem.	Pass (50-64%) The problem and business context are identified and defined but lack a coherent argument or sufficient clarity to justify the relevance and significance of the problem.	Fail (0-49%) Problem and business context are unclear or poorly identified and defined, providing little or no insight into the problem.	2 pts
Data processing (30%) Tasks (ii)-(iv) of Milestone 1 information	High Distinction (85-100%) Quantitative and associated qualitative raw data are exceptionally well processed, including units and uncertainties where relevant. The processed data is presented completely and appropriately and where relevant, includes errors and uncertainties.	Distinction (75-84%) Quantitative and associated qualitative raw data are processed appropriately, including units and uncertainties where relevant. The processed data is presented appropriately and where relevant, includes errors and uncertainties.	Credit (65-74%) Quantitative and associated qualitative raw data are processed, including units and uncertainties where relevant, but with a few mistakes, omissions, and errors. The processed data is presented with some errors and uncertainties.	Pass (50-64%) Quantitative and associated qualitative raw data are processed partially, including units and uncertainties where relevant, but with some mistakes, omissions, and errors. The processed data is presented partially with a few errors.	Fail (0-49%) Quantitative and associated qualitative raw data are processed poorly or to some extent, with mistakes, omissions, and errors. The processed data is presented incomprehensibly and lacks cohesion.	6 pts
Exploratory data analysis (40%) Expected outcomes (ii)-(iii) (See Milestone 1 information)	High Distinction (85-100%) Data are analysed using an extensive and correct range of appropriate codes, models, techniques and/or tools. The analysis is clear, accurate, sophisticated, and comprehensive.	Distinction (75-84%) Data are analysed using a broad and correct range of appropriate codes, models, techniques and/or tools. The analysis is reasonable accurate, well-explained, and coherent.	Credit (65-74%) Data are analysed using a good and reasonable range of appropriate codes, models, techniques and/or tools. The analysis is mostly accurate, and the problem is evident; however not well explained or poorly stated.	Pass (50-64%) Data are analysed using a basic range of appropriate codes, models, techniques and/or tools. The analysis lacks depth and criticality. Poor or no connections between data analysis and the problem are evident.	Fail (0-49%) Data analysis is incorrect and/or inappropriate codes, models, techniques and/or tools used in the analysis are insufficient and/or irrelevant. The analysis is superficial, inaccurate, and/or insufficient.	8 pts
Conclusions and preliminary recommendations (10%) Expected outcome (iii) (iv) (See Milestone 1 information)	High Distinction (85-100%) A detailed conclusion is described and justified, which shows a profound understanding of the problem and business. The data and analysis clearly and critically support the findings. An extensive range of potential recommendations, including some original/innovative/creative solutions, are identified and evaluated using the most relevant and appropriate contextual factors (e.g., constraints, limitations, risks, implications). An extensive range of quantitative/visual information is provided correctly and significantly enhances the report's effectiveness. The recommended solutions leading to the recommendations are original, innovative and creative.	Distinction (75-84%) A conclusion is well-described and justified. It is relevant to the problem and business. The data and analysis support the conclusions. A broad range of potential recommendations are identified and evaluated using most relevant and appropriate contextual factors (e.g., constraints, limitations, risks, implications). The recommended solutions leading to the recommendations are robust, feasible and clearly distinguished.	Credit (65-74%) A conclusion is provided but is only partially relevant to the problem and business. The data and analysis partially support the conclusions. A range of potential recommendations are identified and evaluated using the key relevant and appropriate contextual factors (e.g., constraints, limitations, risks, implications). The recommended solutions are feasible and appropriate.	Pass (50-64%) A conclusion is provided but provides limited relevance to the problem and business. The data and analysis only slightly support the conclusions. A limited range of potential recommendations are identified and evaluated using the relevant and appropriate contextual factors (e.g., constraints, limitations, risks, implications). The recommended solutions are limited and potentially feasible.	Fail (0-49%) A conclusion is irrelevant to the problem and context, and not supported by the data and analysis. The potential recommendations are too basic and lack depth. They are not evaluated referencing relevant and appropriate contextual factors (e.g., constraints, limitations, risks, implications). The recommended solutions are unclear, unfeasible and/or inappropriate.	2 pts
Professional presentation of the report-written/visual (10%)	High Distinction (85-100%) The presentation of report is delivered professionally and engagingly with excellent fluency. Text is accurately formatted, proofread and edited, e.g., spelling, word choice/use of terminology, grammar, referencing, and appropriate length. An extensive range of quantitative/visual information is provided correctly and significantly enhances the report's effectiveness. The relationship between text, visuals and report structure is clear throughout.	Distinction (75-84%) The presentation of report is delivered professionally and engagingly. Text is properly formatted, proofread and edited, e.g., spelling, word choice/use of terminology, grammar, referencing, and appropriate length. A broader and informative range of quantitative/visual information is provided correctly and enhances the report's effectiveness. The relationship between text, visuals and report structure is professional and coherent.	Credit (65-74%) The presentation of report is delivered professionally. Text is mostly properly formatted, proofread and edited, e.g., spelling, word choice/use of terminology, grammar, referencing, and appropriate length. Necessary quantitative/visual information is mostly provided; such information is correct and relevant. The relationship between text, visuals and report structure is generally made coherent.	Pass (50-64%) The presentation of the report is delivered. Text is not consistently fluent and clear. The report has some proofreading and editing issues, including spelling, word choice, use of terminology, grammar, referencing and/or inappropriate length. Quantitative/visual information is limited. The relationship between text, visuals and report structure is mostly clear.	Fail (0-49%) The presentation of the report is poorly delivered. Text is inadequately formatted, proofread and/or edited. The issues can include spelling, word choice, use of terminology, grammar, referencing and/or inappropriate length. Quantitative/visual information is poorly presented, irrelevant and/or incorrect. The relationship between text, visuals and report structure is incoherent.	2 pts